

Adaptive Conformal Prediction for 5G Link Adaptation under Distribution Shift

Emanuele Minotti MINOTTI@KTH.SE and **Genghua Dong** GENGHUA@KTH.SE
KTH Royal Institute of Technology, Stockholm, Sweden

George Koudouridis GEORGE.KOUDOURIDIS@ERICSSON.COM
Ericsson AB, Stockholm, Sweden & AUTH Aristotle University of Thessaloniki, Greece

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Abstract

In 5G link adaptation a modulation and coding scheme (MCS) is chosen under a 10% block-error-rate (BLER) constraint that channel non-stationarity undermines by breaking exchangeability. We drive the significance level α_t from a Conformal Test Martingale (CTM) used as a *continuous control signal* rather than a binary alarm: a windowed mixture martingale tracks non-exchangeability online and a bounded sigmoidal law maps its wealth to α_t . On real traces this restores BLER validity, matches error-counting baselines and—unlike them—returns to its nominal operating point once transient drift subsides.

Keywords: conformal prediction, conformal test martingales, distribution shift, link adaptation, 5G

1. Introduction

A 5G base station selects an MCS maximising spectral efficiency under a 10% BLER target (3GPP, 2018; Martin-Vega et al., 2021). CP (Vovk et al., 2005; Cohen et al., 2023) can wrap a learned, uncalibrated MCS predictor (Tsipi et al., 2024; Sheik Mamode and Fowdur, 2026) in a coverage guarantee—but only under exchangeability, which a non-stationary channel breaks.

CTMs test exchangeability online by betting against p -value uniformity (Vovk et al., 2003; Vovk, 2021), yet serve mainly as *detectors*—triggering retraining (Vovk et al., 2021) or marking change points (Volkhonskiy et al., 2017)—while adaptive CP tunes α_t from coverage errors alone (Gibbs and Candès, 2021; Gibbs and Candès, 2024; Angelopoulos et al., 2023): one family diagnoses shift without acting, the other acts without testing non-exchangeability. We bridge them, driving α_t from CTM wealth.

2. The CTM-Driven Adaptive Framework

From coverage to BLER. We wrap a random-forest MCS classifier with inductive CP (Papadopoulos et al., 2002) and Adaptive Prediction Sets (Romano et al., 2020), picking $m_{\text{sel}} = \min \Gamma^\alpha(x)$. As BLER is monotone in the index, $\alpha = \text{BLER}_{\text{target}}$ maps coverage onto the BLER constraint.

Windowed mixture martingale. To keep the signal bounded and reversible we restrict the product to the W most recent p -values and mixture-average over a grid $\mathcal{E}$ of linear betting functions: $M_t^{(W)} = \frac{1}{|\mathcal{E}|} \sum_{\epsilon \in \mathcal{E}} \prod_{s=\max(1, t-W+1)}^t (1 + \epsilon(p_s - \frac{1}{2}))$. The wealth rises under recent non-uniformity and decays as the window passes the drift: self-releasing, forgoing Ville’s anytime-validity, without a binary reset.

Sigmoidal control law. Once $\log_{10} M_{t-1}^{(W)}$ crosses a threshold $L_{\text{threshold}}$, a sigmoid lowers α_t smoothly from $\alpha_{\text{target}} = 0.10$ toward a floor $\alpha_{\text{target}} - \delta$: a law that is *bounded*, *proportional* in accumulated evidence and *unidirectional* ($\alpha_t \leq \alpha_{\text{target}}$, never under-protected). As $L_{\text{threshold}}$ sits above the martingale’s stationary null envelope, the wrapper is a strict no-op absent drift.

3. Experiments and Results

Data and labels. We use the H-Bahn subset of the open DoNext dataset (Schippers et al., 2025): per-second serving-cell measurements from a public network; chronological 50/20/30% splits, 141,891 test samples. Lacking decoding outcomes, we label via an effective-SINR link-performance model (Brueninghaus et al., 2005; Ikuno et al., 2010) calibrated to the 29 MCS of TS 38.214: the highest-throughput MCS with BLER $\leq 10\%$. Online only the ACK/NACK of the *transmitted* MCS is observed, so this label is counterfactual; a deployment would score the observed outcome, or a covariate-only measure—enough here, as our shifts leave $P(Y | X)$ invariant.

Pipeline and baselines. The windowed martingale uses $W = 1,000$ over a 50-point ϵ -grid. Per-window SINR offsets over ten test windows (labels recomputed; a $P(X)$ shift that still breaks exchangeability) give *gradual* (−6 dB ramp), *abrupt* (−4 dB step) and *transient* (−4 dB pulse) scenarios. Six methods share the ICP infrastructure and differ only in how α_t is set—bare RF, static ICP, ACI (Gibbs and Candes, 2021), DtACI (Gibbs and Candès, 2024), Conformal PID (Angelopoulos et al., 2023), ours.

Validity is restored. Under drift the bare predictor’s BLER violation rate explodes; *every* CP method drives it below 10%. Ranking the $k = 6$ methods within each of the ten test windows of the three scenarios— $N = 30$ blocks—a Friedman–Nemenyi test (Demšar, 2006) finds only the bare RF significantly worse; among the five CP methods no pairwise difference is significant. Blocks within a scenario are temporally dependent, so near-critical verdicts are soft, and validity here is empirical: finite-sample bounds for data-dependent α_t remain open.

The correction self-releases. Transient drift is decisive: once the pulse leaves the window the wealth decays, α_t returns to 0.10, and throughput to its stationary, static-ICP level—no residue. The baselines never return to a nominal level: their α_t is 0.13–0.17 *before* the drift and 0.16–0.18 after, tracking coverage slack accumulated by the conservative $\min \Gamma^\alpha$ policy, not the channel. As larger α_t means smaller sets and a higher MCS, that slack buys them *more* throughput, not more caution: what they lack is not conservatism but a readable drift signal.

Conclusion. CTM wealth thus acts as a controller, not merely a detector: reliability equal to the baselines, uniquely released after transient drift, at a 4–13 pp throughput cost.

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